

Umesh Mahind

AI/ML Engineer — M.E., Ph.D*

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Summary

AI/ML Engineer with 9+ years of experience designing and delivering production-grade Generative AI, Machine Learning, and NLP solutions across Manufacturing, Oil & Gas, Healthcare, Insurance, and Finance domains. Led cross-functional AI teams to architect, develop, and deploy enterprise AI applications. Expertise in LLMs, RAG, Agentic AI, MLOps, and cloud-native AI solutions on Azure and AWS, delivering measurable business impact.

Skills

GenAI & NLP:	LLMs (GPT-4.1, GPT-5, Claude 3.5), LangChain, RAG, Prompt Engineering, Agentic AI, Transformers, FAISS, Chroma
Machine Learning:	Regression, Classification, Clustering, XGBoost, LightGBM, Time Series Forecasting, Statistical Analysis, Model Evaluation
Deep Learning:	TensorFlow, PyTorch, Keras, CNN, RNN, LSTM
Programming:	Python, SQL
Cloud:	AWS SageMaker, Amazon S3, AWS Lambda, EC2, CloudWatch, GCP, Azure OpenAI, Azure AI Search, Azure Blob Storage, Azure Functions
MLOps/LLMOps:	Docker, FastAPI, Flask, Streamlit, MLflow, CI/CD, Prompt Evaluation, Hallucination Detection, Guardrails, Model Monitoring
Data Tools:	Pandas, NumPy, Snowflake, Airflow
Visualization:	Power BI, Tableau, Matplotlib

Experience

Accenture Solutions Pvt. Ltd., Mumbai

Aug 2024 – Present

AI/ML Associate Manager

• Caterpillar (Manufacturing — Agentic AI/GenAI Projects):

- Designed and deployed a **Generative AI** PRU Quality Gate solution using **Azure OpenAI**, Azure AI Search, Azure Blob Storage, FastAPI, and Docker for automated engineering drawing validation, improving quality performance by **85%** and saving **2250+ engineering hours annually**.
- Built an enterprise **Generative AI** Product Configuration Automation platform using **Azure OpenAI**, **LangChain**, **Agentic AI**, FastAPI, and Docker to process engineering documents (PAN), reducing manual effort and delivering **\$50K–\$100K** annual cost savings.
- Developed an **LLM**-powered Quality Insights platform using **Azure OpenAI**, **Claude 3.5**, **LangChain**, **RAG**, and Snowflake to summarize quality events, identify recurring manufacturing issues, and improve engineering decision-making.

• IOCL (Energy — ML & Analytics Projects):

- Developed scalable **Machine Learning** pipelines on **AWS SageMaker**, Amazon S3, AWS Lambda, and **MLflow** using **XGBoost** and **LightGBM**, improving sales forecasting accuracy by **20%**.
- Built **Time Series Forecasting** models using **AWS SageMaker**, Python, and Scikit-learn, achieving **95%** prediction accuracy for demand planning.
- Designed a **Customer Churn Prediction** solution using **AWS SageMaker**, **MLflow**, and clustering techniques, achieving **85%** prediction accuracy.

- Performed large-scale customer analytics using **AWS SageMaker**, SQL, and Python, identifying approximately **20%** revenue leakage and supporting **Revenue Growth Management (RGM)** initiatives.

LexisNexis Risk Solutions, Relx India Pvt. Ltd., Mumbai
Software Engineer

Jul 2022 – Aug 2024

- Built real-time anomaly detection system reducing manual effort by 60%.
- Improved ML model performance by 25% via feature engineering.
- Developed Power BI dashboards for business insights.

Aditya Birla Health Insurance, Mumbai
Data Scientist / RPA Developer

Jan 2022 – Jul 2022

- Automated Excel workflows reducing manual effort by 80% and saving 20+ hours/week.
- Built ML models improving prediction accuracy by 25%.

UNIT, SNDT University, Mumbai
Assistant Professor & Data Science Project Mentor

Aug 2018 – Jul 2021

- Mentored students on Machine Learning and Data Science projects
- Designed and delivered curriculum on Python, statistics, and ML concepts with practical industry use cases.

SIES Graduate School of Technology (GST), Mumbai
Python Lecturer

Aug 2015 – Aug 2017

- Delivered Python programming training covering core concepts

Education

Ph.D.	University of Mumbai (Thesis Submitted)	Pursuing
M.E.	University of Mumbai (CGPI: 7.80)	2014 – 2016
B.E.	University of Mumbai (60.12%)	2011 – 2014
Diploma	M.S.B.T.E. (78.91%)	2007 – 2010

Certifications

- **Anthropic:** Claude Certified Architect-Foundation (2026)
- **Accenture:** Reinvention with Agentic AI – Facilitator (2026)
- **Udemy:** MLOps (AWS, GCP) (2026), LLM Engineering (2025)
- **NVIDIA:** Building LLM Applications (2025)
- **Google:** Machine Learning Certification (2025)
- **DeepLearning.AI:** Neural Networks and Deep Learning (2025)
- **Coursera:** Prompt Engineering (2025)

Research & Contributions

- Filed 1 patent in AI/ML domain related to intelligent automation and predictive analytics.
- Published 10+ research papers in reputed journals and international conferences focusing on Machine Learning, NLP, and Data Science.
- Delivered 10+ expert talks and technical sessions on AI/ML, Generative AI, and data-driven systems.